

Homework 1¹

Due date: September 15, 2026 (Tuesday).

Please submit your answer by 11:59pm.

There are a total of 3 questions. Attach your R code.

Q1 (Basic R questions):

1. If you are new to R, go over the following material:

<https://www.zeileis.org/teaching/AER/Ch-Basics.pdf>

There will be R DataCamp assignments throughout the semester to deepen your understanding.

2. Write an R script that ...

- (a) Create the following two matrices

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$$

- (b) Compute

$$C = A^{-1}B$$

- (c) Print out C' on your screen (i.e., transpose of C)

- (d) Explain the difference between `A %*% B` and `A*B`.

3. Write another R script that computes the following quantity using a `for` loop

$$X = \sum_{k=1}^{100} D_k \quad \text{where } D_k = \begin{pmatrix} k & k+1 \\ k^2 & k/2 \end{pmatrix}.$$

Q2 (Simulation question): Read [labnotes01.pdf](#) and solve the quiz at the end of the notes. You can find notes and associated Rmd files from Canvas:

... > Files > Data sets and labs > lab 1

¹Last compiled: August 31, 2026; STAT5200

Q3 (Paper questions): Find and read a paper that uses econometric methods

1. Pick a field of your interest. If you don't have any, let it be "economics".
2. List five academic journals that are well-read in the field of your interest. The easiest way (which may not be perfect, though) is to look at Google Scholar ranking:
https://scholar.google.com/citations?view_op=top_venues&hl=en
3. Find a paper that (1) was published in one of those journals, AND (2) contains one of the following words

econometric, econometric method(s), econometric technique(s),

AND (3) applies an econometric method to data, AND (4) presents results from their analysis.

One way to find such a paper is to use Google Scholar. Type the following in the search box

`source:"[name of the journal]" "econometric"`

4. Properly cite the paper you found (name of the author, the title of the article, year of publication, the name of the journal, etc.)
5. Read the paper and explain the main research question of the paper in one paragraph.
6. What kind of econometric method do the authors use? You don't need to understand the method. Just list the name of the method.
7. When the authors present their empirical findings, do they report standard errors?
8. (This question will NOT be graded) You don't have to know this yet, but can you explain what the standard error is?